1.Water Jug Problem

Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

Problem

You are given two jugs:

- Jug A = 4 gallons
- Jug B = 3 gallons

There is a water pump, but there are no measuring marks on the jugs.

Allowed operations:

- Fill a jug completely.
- Empty a jug completely.
- Pour water from one jug to another until one jug becomes full or the other becomes empty.

Find a way to get exactly 2 gallons in the 4-gallon jug.

State Representation

A state is represented as (A, B) where:

- A = amount of water in the 4-gallon jug.
- B = amount of water in the 3-gallon jug.

Example:

- (4,0) → 4-gallon jug is full and 3-gallon jug is empty.
- (2,3) → 4-gallon jug has 2 gallons and 3-gallon jug is full.

Algorithm

1. Fill the 3-gallon jug.
2. Pour the water into the 4-gallon jug.
3. Fill the 3-gallon jug again.
4. Pour water into the 4-gallon jug until it becomes full.
5. Empty the 4-gallon jug.

6. Pour the remaining water from the 3-gallon jug into the 4-gallon jug.
7. Fill the 3-gallon jug again.
8. Pour it into the 4-gallon jug.
9. The 4-gallon jug now contains exactly 2 gallons.

Pseudocode

START

Fill Jug B (3 gallons)

Pour Jug B into Jug A

Fill Jug B again

Pour Jug B into Jug A until Jug A is full

Empty Jug A

Pour remaining water from Jug B into Jug A

Fill Jug B

Pour Jug B into Jug A

STOP

Step-by-Step Solution

Step Action State (4-gal,3-gal) Initial Both empty (0,0)

1 Fill 3-gallon jug (0,3)

2 Pour into 4-gallon jug (3,0)

3 Fill 3-gallon jug again (3,3)

4 Pour into 4-gallon jug until full (4,2)

5 Empty 4-gallon jug (0,2)

6 Pour remaining 2 gallons into 4-gallon jug (2,0)

Final Answer

The 4-gallon jug contains exactly 2 gallons, which satisfies the requirement.

2. Mars Rover Agent

Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

i. What are the percept's for this agent?
ii. Characterize the operating environment.
iii. What are the actions the agent can take?
iv. How can one evaluate the performance of the agent?
v. What sort of agent architecture do you think is most suitable for this agent? Why?

Problem

The Mars Rover is an intelligent robotic agent that travels on the surface of Mars. It collects rock and soil samples, captures images, analyzes the environment, and sends information back to Earth without continuous human control.

(i) What are the percepts for this agent?

Percepts are the information received by the rover through its sensors.

The Mars Rover receives:

- Images from cameras
- Rock and soil information
- Temperature
- Atmospheric pressure
- Light intensity
- Terrain (rocks, slopes, sand)
- Position and direction
- Obstacle detection
- Battery level

(ii) Characterize the operating environment

Environment Property Description

Observable Partially Observable (cannot see everything around it) Agents

Single Agent

Deterministic Stochastic (environment can change unexpectedly)

Episodic/Sequential Sequential
Environment Property Description

Static/Dynamic Dynamic

Discrete/Continuous Continuous

Known/Unknown Partially Unknown

(iii) What actions can the agent take?

The Mars Rover can:

- Move forward
- Move backward
- Turn left
- Turn right
- Avoid obstacles
- Capture images
- Drill rocks
- Collect soil samples
- Analyze samples
- Measure temperature
- Send data to Earth
- Recharge using solar panels
- Stop when battery is low

(iv) How can one evaluate the performance of the agent?

The performance of the Mars Rover is evaluated based on:

● Number of successful samples collected ● Accuracy of

scientific analysis

- Distance travelled safely
- Number of obstacles avoided
- Battery efficiency
- Amount of useful data sent to Earth
- Mission completion time
- Successful completion of mission objectives

A rover that collects more accurate samples, avoids hazards, and completes its mission safely is considered to have better performance.

(v) What sort of agent architecture is most suitable? Why?

The most suitable architecture is a Utility-Based Learning Agent.

Why?

- It learns from previous experiences.
- It chooses the safest and most efficient path.
- It can work even when communication with Earth is delayed.
- It adapts to unknown environments.
- It maximizes scientific data collection while minimizing risks and power consumption.

Simple Working of Mars Rover Agent

Sensors

↓

Perceive Environment

↓

Analyze Data

↓

Select Best Action

↓

Move / Collect Sample / Capture Image

↓

Store & Send Data to Earth

↓

Repeat Until Mission Ends

Result

The Mars Rover acts as an intelligent autonomous agent. It observes the Martian environment using sensors, makes decisions based on the collected information, performs scientific tasks, and sends valuable data to Earth while operating safely and efficiently.

3. 8-Queens Problem

Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none

of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

Problem

Place 8 queens on an 8 × 8 chessboard such that no two queens attack each

other. A queen can attack:

● Horizontally

● Vertically

● Diagonally

The goal is to place all 8 queens so that none of them are in the same row, column, or diagonal.

Problem Formulation

● Initial State: Empty 8 × 8 chessboard.

● State Space: All possible arrangements of queens on the board.

● Operators: Place one queen in a safe position.

● Goal State: Eight queens are placed without attacking each other.

● Path Cost: Each move has equal cost, so the path cost is not important.

Search Strategy

The Backtracking Search Algorithm is commonly used.

Working

1. Start from the first column.

2. Place a queen in a safe row.

3. Move to the next column.

4. If no safe position is available, go back (backtrack) to the previous column.

5. Try another position.

6. Continue until all 8 queens are placed.

Algorithm

1. Place the first queen in the first column.

2. Check whether the position is safe. 3. If safe, place the queen.

4. Move to the next column.

5. If no safe position exists, backtrack. 6. Repeat until all 8 queens are placed.

Pseudocode

```
START
PlaceQueen(column)
IF column = 8
   Print Solution
   RETURN TRUE
FOR each row from 0 to 7
   IF position is safe
      Place Queen
      IF PlaceQueen(column + 1)
         RETURN TRUE
      Remove Queen
RETURN FALSE
STOP
```

Example Solution

One valid arrangement is:

Column Row

1 1

2 5

3 8

4 6

5 3

Column Row

6 7

7 2

8 4

No two queens attack each other in this arrangement.

Advantages

- Finds a correct solution efficiently.
- Removes unnecessary search paths.
- Suitable for constraint satisfaction problems.

Result

Thus, the 8-Queens Problem is solved using the Backtracking Search Strategy, where queens are placed one by one while ensuring that no two queens attack each other.

4. OLA Cab Booking Application

A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

Problem

A customer wants to travel from one location to another using the OLA Cab Booking mobile application. The customer can choose different cab types such as Mini, Micro, Sedan, Shared, Prime, etc., based on comfort and availability.

Type of Problem-Solving Agent

The most suitable agent is a Goal-Based Agent.

Why?

- The customer's goal is to reach the destination.
- The agent compares available cab options.
- It selects the cab that best satisfies the user's preferences, such as comfort, cost, and availability.

- After selecting the best option, it books the cab to achieve the goal.

Working of the Goal-Based Agent

1. Receive pickup and destination.
2. Search for available cabs.
3. Display different cab types.
4. Compare fare and arrival time.
5. Customer selects a cab.
6. Confirm booking.
7. Track the ride until the destination is reached.

Algorithm

1. Start the application.
2. Enter pickup and destination.
3. Find available cabs.
4. Display cab types with fare and estimated arrival time.
5. Customer selects a cab.
6. Confirm booking.
7. Driver is assigned.
8. Track the trip.
9. End the ride after reaching the destination.

Pseudocode

START

Input Pickup Location

Input Destination

Search Available Cabs

Display Cab Types

(Mini, Micro, Sedan, Shared, Prime)

Customer Selects Cab

IF Cab Available THEN

Confirm Booking

Assign Driver

Start Trip

Reach Destination

Complete Payment

ELSE

Display "No Cab Available"

ENDIF

STOP

Flow of the OLA Booking

Process Customer Opens App

↓

Enter Pickup & Destination

↓

Search Available Cabs

↓

Display Fare & Cab Types ↓

Customer Selects Cab

↓

Driver Assigned

↓

Trip Starts

↓

Reach Destination

↓

Payment Completed

Advantages

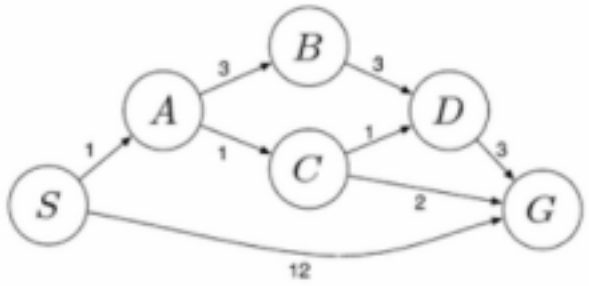
- Provides multiple cab options.
- Helps users choose based on cost and comfort.
- Reduces waiting time.
- Makes travel simple and convenient.

Result

Thus, the Goal-Based Agent is the most suitable problem-solving agent for the OLA Cab Booking application because it selects the best cab option to achieve the customer's goal of reaching the destination safely and efficiently.

5. Uniform Cost Search (UCS) – Logistics Delivery Network

A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm. The delivery network is represented as a weighted graph:



Problem

A logistics company wants to transport a package from the starting warehouse S to the destination warehouse G with the minimum travel cost. The delivery network is represented as a weighted graph, where each edge represents the travel cost (fuel + time). The task is to find the least-cost path using the Uniform Cost Search (UCS) algorithm.

Graph Details

From the given graph:

Route Cost

S → A 1
Route Cost

S → G 12

A → B 3

A → C 1

B → D 3

C → D 1

C → G 2

D → G 3

Problem Formulation

● Initial State: Warehouse S

● Goal State: Warehouse G

● State Space: All warehouses (S, A, B, C, D, G)

● Operators: Move from one warehouse to another through connected routes.

● Path Cost: Sum of all edge costs from the start to the destination.

Method Used

Uniform Cost Search (UCS)

UCS always expands the node with the lowest cumulative cost first. It guarantees the optimal (least-cost) path when all edge costs are positive.

Algorithm

1. Start from node S with cost 0.
2. Add the start node to the priority queue.
3. Remove the node having the smallest cumulative cost.
4. Expand its neighboring nodes.
5. Update their cumulative costs.
6. Continue until the goal node G is reached.
7. Return the least-cost path and total cost.

Pseudocode
START

Insert Start Node S with cost 0 into Priority Queue

WHILE Queue is not empty

Remove node with minimum cost

IF Goal Node is reached

Return Path and Cost

Expand all neighboring nodes

Add neighbors with updated cumulative cost

END WHILE

STOP

UCS Search Process

Step 1

Start at S

Node Cost

S 0

Expand S

Queue:

- A (1)
- G (12)

Step 2

Choose the smallest cost.

Expand A (1)

New paths:

- B = 1 + 3 = 4

- C = 1 + 1 = 2

Queue:

- C (2)
- B (4)
- G (12)

Step 3

Expand C (2)

New paths:

- D = 2 + 1 = 3 ●

G = 2 + 2 = 4 Queue:

- D (3)
- B (4)
- G (4)
- G (12)

Step 4

Expand D (3)

New path:

- G = 3 + 3 = 6

Queue:

- B (4)
- G (4)
- G (6)
- G (12)

Step 5

Expand G (4)

Goal node reached.

Cost Calculation

Possible paths:

Path Total Cost

S → G 12

S → A → B → D →

G 1 + 3 + 3 + 3 = 10

S → A → C → D →

G 1 + 1 + 1 + 3 = 6

S → A → C → G 1 + 1 + 2 = 4 ✅

Least-Cost Path

S → A → C → G

Total Cost = 4

Interpretation of Result

Uniform Cost Search compares all possible routes based on their cumulative costs and always expands the lowest-cost path first. From the given graph, the path S → A → C → G has the minimum total cost of 4, making it the most efficient route for transporting the package. This reduces both travel time and fuel cost, making it suitable for logistics and delivery applications.

Result

Thus, by applying the Uniform Cost Search (UCS) algorithm, the least-cost path from S to G is:

S → A → C → G

Minimum Cost = 4